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Railway applications - Track - Acceptance of works - Part 3: Acceptance of rail grinding, milling and planing work in track

Applications ferroviaires - Voie - Réception des travaux - Partie 3 : Critères de réception des travaux de meulage, fraisage et rabotage des rails en voie

Bahnanwendungen - Oberbau - Abnahme von Arbeiten - Teil 3: Abnahme von Schleif-, Fräs- und Hobelarbeiten an Schienen im Gleis

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Foreword

This document (prEN 13231-3:2006) has been prepared by Technical Committee CEN/TC 256 “Railway applications”, the secretariat of which is held by DIN.

This document is currently submitted to the Formal Vote.

This European Standard is one of the series EN 13231 "*Railway applications – Track – Acceptance of works*" as listed below:

— *Part 1: Works on ballasted track - Plain line*

— *Part 2: Works on ballasted track - Switches and crossings*

— *Part 3: Works on ballasted track – Acceptance of rail grinding, milling and planing work in track*

1 Scope

This part of this European Standard lays down the technical requirements and the measurements to be made for the acceptance of work to re-profile both longitudinally and transversely the heads of railway rails, including the parts of switches and crossings that can be reprofiled.

For acceptance purposes, two classes of longitudinal profile and three classes of transverse profile tolerance are defined.

It also informs about procedures to verify reference instruments to be used for these measurements and informs about a method to approve non-reference instruments to be used for measurements.

It applies to reprofiled vignole railway rails 40 kg/m and above.

A form of acceptance documentation that may be used is given in Annex C.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN ISO 3274, *Geometrical product specifications (GPS) – Surface texture: Profile method – Nominal characteristics of contact (stylus) instruments (ISO 3274:1996)*

EN ISO 4287, *Geometrical product specifications (GPS) – Surface texture: Profile method – Terms, definitions and surface texture parameters (ISO 4287:1997)*

EN ISO 4288, *Geometrical product specifications (GPS) – Surface texture: Profile method – Rules and procedures for the assessment of surface texture (ISO 4288:1996)*

EN ISO 11562:1997, *Geometrical product specifications (GPS) – Surface texture: Profile method – Metrological characteristics of phase correct filters (ISO 11562:1996)*

ISO 3611, *Micrometer callipers for external measurement*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1
reference profile
the transverse profile to which rail is to be reprofiled, within the specified tolerances

3.2
reference line
the line normal to the track's longitudinal axis and tangent to the heads of both rails

3.3
angle of inclination of rail
the nominal angle at which rail is laid (see Figure 1(b)) e.g. 0° (vertical rails), 2,86° (1:20 inclination), 1,91° (1:30 inclination) etc., inclined towards the centre of the track

NOTE For rail which is laid in non-canted track, the angle of inclination of the rail is equal to the angle between the vertical and the centre-line of the inclined rail.

3.4
rail crown
that point on the rail-head surface that is aligned with the centre-line of the web

3.5
reference rail
a rail with the reference profile, at the desired angle of inclination relative to the reference line (see Figure 1(a))

NOTE In switch and crossing work, with rails layed vertically, it may be desired to reprofile the rail so that its head profile matches that of the adjacent plain line in which rails are inclined at, for example, 1 in 20. In this case, the angle of inclination of the reference rail will be 2,86° (1 in 20).

3.6
reference point A
that point towards the gauge side of a reference rail at which the angle between the reference line and the tangent to the profile is equal to the specified angle of inclination (see Figure 1)

3.7
reference point B_1
that point on the gauge face of a reference rail which lies 14 mm below that line that is parallel to the reference line and which passes through reference point A (see Figure 1(a))

3.8
reference point B_2
that point on the gauge corner of a reference rail at which a line which is tangent to the rail lies at an angle of 45° to the reference line (see Figure 1(b))

3.9
reprofiling zone
that area of the railhead of a reference rail between the point at which the tangent to the rail lies at an angle of 70° to the reference line, measured towards the gauge side of the rail, and the point at which the tangent to the rail lies at an angle of 5° to the reference line, measured towards the field side of the rail (see Figure 2)

3.10
reference instrument
an instrument for the measurement of longitudinal or transverse profile the performance of which has been verified in accordance with the procedure agreed upon between the contractor and the customer (an example is given in Annex A)

3.11
approved instrument
an instrument for measurement of longitudinal or transverse profile the usage of which is justified by correlation of its performance with that of a reference instrument in accordance with the agreement between the contractor and the customer (an example is given in Annex B)

3.12
test instrument
an instrument whose use as a reference instrument or an approved instrument is being tested

3.13
reprofiling site
a continuous length of track where the rail is to be reprofiled

3.14
grinding facet
an approximately plane sector of the profile of a rail reprofiled by grinding, which is produced by a single grinding stone whose axis of rotation is normal to the rail's longitudinal axis

3.15
characteristic length
the length on the rail travelled during one rotation of a grinding stone

3.16
traced profile
the profile of the rail as recorded by the measuring system

3.17
profile filter
an electronic device which separates profiles into long-wave and short-wave components, or into components within a specified wavelength range

3.18
phase correct profile filter
profile filter which does not cause phase shifts which lead to asymmetrical profile distortions (see EN ISO 11562:1997)

3.19
cut-off wavelength
wavelength of a sinusoidal profile of which 50 % of the amplitude is transmitted by the profile filter

NOTE Profile filters are identified by their cut-off wavelength value (see EN ISO 11562).

3.20
filtered profile
the profile which results from applying a profile filter to the primary profile

3.21
window
that section of a record of longitudinal profile within which an average amplitude of the record is calculated, and whose length is specified

3.22
moving average of peak-to-peak amplitudes
the average depth of individual irregularities in a profile which is calculated as a quasi-continuous function of distance along the track; the depth is calculated for that section of the profile which lies within a window of specified length (see Annex D); the function with distance is calculated by incrementing the window along the profile by a distance equal to the sampling interval

3.23
moving average of root-mean-square (RMS) amplitudes
the root-mean-square amplitude of a profile which is calculated as a quasi-continuous function of distance along the track; the average is calculated for that section of the profile which lies within a window of specified length (see Annex D); the function with distance is calculated by incrementing the window along the profile by a distance equal to the sampling interval

3.24
transition length
initial or final section of a length of track where the validity of a measurement of longitudinal or transverse profile is questionable for a variety of reasons, including settling of electronic and digital components and circuits

3.25
sampling interval
the distance between successive points at which a continuous record of the traced profile is sampled in order to produce the primary profile

3.26
primary profile
representation of the measured longitudinal profile before application of any profile filter

3.27
percentage exceedance
the percentage length of a test site over which a measurement of the amplitude of the filtered profile exceeds a prescribed limit

3.28
deviation of the measured profile
the deviation between the measured profile and the reference rail, measured normal to the surface of the reference rail when the measured profile and the reference rail are aligned at points A and B_1 or A and B_2, without rotation of either profile. The deviation is considered positive when the measured profile is above the reference rail (see Figure 3)

3.29
range of deviation
the difference between the maximum and minimum values of the deviation of the measured profile (see Figure 3)

3.30
class 1, class 2
classes of longitudinal profile differentiated by the proportion of a grinding site reaching a specified standard (see 4.3)

3.31
class Q, class R, class S
classes of transverse profile differentiated by the proportion of a grinding site reaching a specified standard (see 5.3)

4 Longitudinal profile

4.1 Principle

Measurements are made using either a reference instrument (see 3.10) or an approved instrument (see 3.11). Approved instruments do not offer the same accuracy as reference instruments but are generally adequate for the purpose of demonstrating compliance with the requirements of this standard.

NOTE An example of an approved instrument is the type of system used for routine measurements on grinding trains. Some of the systems used for routine measurements on reprofiling trains fall into this category.

In accordance with current practice, limits are set on the magnitude of the irregularities that can remain in track after a reprofiling operation. It is recognised, however, that it can be uneconomic to achieve 100 % compliance with these, particularly where isolated top faults, such as wheelburn, exist prior to reprofiling. Two Classes are therefore offered, differentiated by the percentage of the reprofiled track meeting the specified criteria. Where isolated top faults exist, Class 2 offers a lower cost option compared to Class 1 as it will be achieved with fewer passes. However a larger number of isolated non-compliant zones will remain in the reprofiled site.

Class 1 also includes limits for very short (10 mm – 30 mm) and very long (300 mm – 1 000 mm) wavelength residual irregularities; these are not included in Class 2. Where corrugations in these wavebands are required to be removed it will also be necessary to specify Class 1.

This standard permits the quality of the longitudinal profile to be characterised in any one of the three ways, at the discretion of the client. These are the moving average RMS amplitude of the irregularities in a given waveband, the moving average peak-to-peak amplitude of irregularities in a given waveband and the number of irregularities exceeding the specified limit per 100 m of track. Except in borderline cases all three methods are expected to give similar results.

4.2 Measurements required

The longitudinal profile of the finished reprofiled rail shall be recorded continuously using either a reference instrument or an approved instrument. Where independent verification is required a reference instrument shall be used. All measurements undertaken in order to demonstrate compliance with 4.3 shall be recorded.

NOTE 1 For measurements in the (10 mm to 30 mm) wavelength range, it is at present unlikely that instruments other than reference instruments will have sufficient accuracy.

Measurements shall be undertaken at the latest within 8 days of reprofiling or after the track has carried 0,3 MGT of traffic, whichever occurs sooner.

NOTE 2 It is preferable for measurements to be made immediately after reprofiling.

Longitudinal profile measurements shall be made within a distance of 15 mm laterally on the rail from the rail crown, to produce the traced profile.

NOTE 3 It is recommended that a digital form of the traced profile, the primary profile, be used for subsequent analysis.

4.3 Acceptance criteria for longitudinal profile

4.3.1 General

The acceptance of re-profiled sites shall be on the basis of one of the following sets of criteria, as specified in the contract:

— moving average of RMS amplitudes (see 4.3.2);

— moving average of peak-to-peak amplitudes (see 4.3.3);

— the number of irregularities exceeding a specified amplitude (see 4.3.4).

4.3.2 Moving average of RMS amplitudes

The primary or traced profile shall be processed to provide a filtered profile within each of the wavelength ranges given in Table 1. The cut-off wavelengths for each wavelength range and the length of the corresponding window within which the pertinent moving average is to be calculated are also given in Table 1.

The percentages of any site in which the moving average RMS amplitudes exceed the values specified in Table 2 shall be calculated. These percentages shall not exceed the limits given in Table 4 for the Class specified.

4.3.3 Moving average of peak-to-peak amplitude

The primary or traced profile shall be processed to provide a filtered profile within each of the wavelength ranges given in Table 1. The cut-off wavelengths for each wavelength range and the length of the corresponding window within which the pertinent moving average is to be calculated are also given in Table 1.

The percentage of any site in which the moving average of peak-to-peak amplitude exceeds the value specified in Table 3 shall be calculated. These percentages shall not exceed the limits given in Table 4 for the class specified.

4.3.4 Number of irregularities exceeding a specified amplitude

The primary or traced profile shall be processed to provide a filtered profile within each of the wavelength ranges given in Table 1. The cut-off wavelengths for each wavelength range are also given in Table 1.

Irregularities in any filtered profile, the peak-to-peak amplitude of which exceeds the limits specified in Table 5 shall be identified. The number of irregularities exceeding these limits in any 100 m of rail shall not exceed the limits given in Table 5 for the class specified.

Table 1 — Window lengths

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1 000
Window length (m)	0,15	0,5	1,5	5

Table 2 — Moving average of RMS amplitude limits

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1 000
Limit of moving average of RMS amplitude (mm)	0,004	0,004	0,012	0,040

Table 3 — Moving average of peak-to-peak amplitude limits

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1 000
Limit of moving average of peak-to-peak amplitude (mm)	0,010	0,010	0,030	0,100

Table 4 — Acceptance criteria for longitudinal profile expressed in terms of allowable percentages of track exceeding moving average RMS or peak-to-peak amplitude limits

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1 000
Class 1	5 %	5 %	5 %	10 %
Class 2	No requirement	10 %	10 %	No requirement

Table 5 — Acceptance criteria for longitudinal profile expressed in terms of numbers of exceedances per 100 m of rail

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1 000
Limit on peak-to-peak amplitudes (mm)	0,010	0,010	0,030	0,100
Class 1	250	100	25	10
Class 2	No requirement	200	50	No requirement

5 Transverse profile

5.1 Principle

Measurements are made using either a reference instrument (see 3.10), or an approved (see 3.11) instrument. Approved instruments do not offer the same accuracy as reference instruments but are generally adequate for the purpose of demonstrating compliance with the requirements of this standard.

Reprofiling can be undertaken for a variety of reasons. Where reprofiling is undertaken purely for the removal of corrugation, there may be less need for the rail to be reprofiled with precision. In other cases, it may be necessary for the reprofiled rail to match closely the ideal profile, represented by the reference rail (see 3.5). A range of Classes is therefore included to enable the client to specify the level of precision that is appropriate for the site to be reprofiled.

NOTE Where reprofiling is undertaken to improve conicity, Class Q (see 5.3) is likely to be appropriate.

The match between the reprofiled rail and the profile of the reference rail is determined by aligning the two at two points and measuring maximum difference between them (see Figure 3). For straight track, these points of alignment generally approximate to the rail crown and the gauge point. On the high rail of curves this method is not applicable if sidewear has occurred and an alternative method of alignment is therefore used.

5.2 Measurements required

The rail's transverse profile shall be measured using either a reference instrument or an approved instrument. Where independent verification is required a reference instrument shall be used. All measurements undertaken in order to demonstrate compliance with 5.3 shall be recorded.

Measurements shall be made at the latest within 8 days of reprofiling or before the track has carried 0,3 MGT of traffic, whichever occurs sooner.

NOTE It is preferable for measurements to be made immediately after reprofiling.

The transverse profile of each finished, reprofiled rail shall be measured sufficiently frequently to ensure compliance with the requirements stated in 5.3. The transverse profile shall be recorded at least once per reprofiling site or at an interval of not more than 500 m on a reprofiling site greater than 500 m long.

Where independent verification is required, measurements of each rail shall be made at an interval of not less than 10 m throughout the reprofiling site.

5.3 Acceptance criteria for the transverse profile

Each measured profile shall be aligned with the appropriate reference rail so that the reference points A and B_1, or A and B_2, on the reference rail coincide with points on the measured profile. The alignment shall be undertaken without rotation of either profile.

Reference points A and B_2 shall be used on side-worn rails and A and B_1 elsewhere.

The percentage of measurements in which the range of deviation is less than 0,6 mm, 1,0 mm and 1,7 mm shall be calculated and shall not be less than the values given in Table 6(a) for the Class specified.

Table 6(a) — Minimum proportion of measurements within the specified range

Range of deviation (mm)	0,6	1,0	1,7
Class Q	90 %	95 %	98 %
Class R	75 %	85 %	98 %
Class S	No requirement	No requirement	75 %

The maximum positive and negative deviations shall not exceed those in one of groups A, B or C, as specified in the contract.

Table 6(b) — Maximum permitted deviations

Range of deviation (mm)	**Permissible maximum deviations (mm)**		
	Group A	**Group B**	**Group C**
≤ 0,6	+0,6/ -0	+ 0,3/ - 0,3	+ 0/ - 0,6
> 0,6; ≤ 1,0	+1,0/ -0	+ 0,5/ - 0,5	+ 0/ - 1,0
> 1,0; ≤ 1,7	+1,7/ -0	+ 0,8/ - 0,9	+ 0/ - 1,7

6 Metal removal

6.1 Measurements required

Measurements of metal removal from the rail-head are required only if there is a requirement in the contract to demonstrate a minimum or maximum depth of metal removal. All measurements undertaken in order to demonstrate compliance with 6.2 shall be recorded.

The height of the rail shall be measured using a micrometer whose accuracy is in accordance with ISO 3611.

The rail depth or depth of the rail-head shall be measured before and after reprofiling at a minimum of 5 positions on each rail at distance of no less than 0,5 m apart. Measurements shall be made within a month of reprofiling. The rail shall be marked to ensure that measurements before and after reprofiling are made within a distance of 10 mm of each other along the rail. If the rail is initially corrugated, measurements shall be undertaken in the trough of the corrugation.

Measurement of the rail height or depth of the rail-head shall be processed so as to provide the depth of metal removed within 15 mm transversely of the rail crown, or elsewhere on the rail-head as agreed between client and contractor.

NOTE Measurements should be recorded once per week or as required by the contract.

6.2 Acceptance criteria for metal removal

The minimum or maximum depth of metal removed shall be as required by the contract at the minimum percentage of measurement positions specified by the contract.

7 Surface roughness

Roughness shall be measured using an instrument that complies with EN ISO 3274.

NOTE 1 Alternative instruments may be used for production control processes.

Measurements shall be made immediately after reprofiling along the rail's longitudinal axis within 15 mm of the rail crown. Where applicable, measurements shall be made on the same grinding facet. A roughness sampling length, lr = 2,5 mm and a roughness evaluation length, ln = 12,5 mm, as defined in EN ISO 4288, shall be used.

At least 6 contiguous measurements shall be taken. Where applicable, sufficient measurements shall be made to cover at least one complete characteristic length on the grinding facet.

The arithmetic mean surface roughness (R_a), as defined in EN ISO 4287, of each measured section, shall be calculated.

Unless specified otherwise in the contract, in no more than 16 % of the measured lengths (or 1 in 6, if only 6 measurements are made) shall R_a exceed 10 µm.

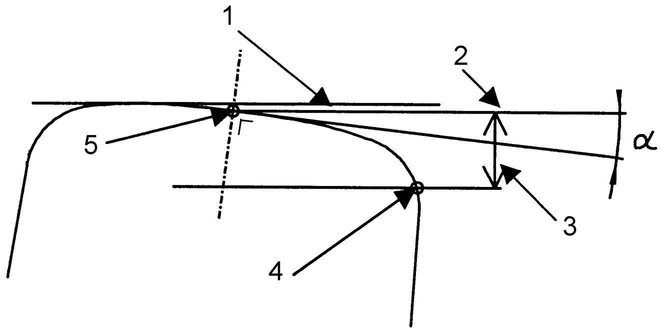
NOTE 2 Other values of surface roughness may be required where reprofiling is undertaken for the purpose of noise abatement or the improvement of adhesion.

8 Visual appearance: acceptance criteria

Where facets are produced by the re-profiling operation, the maximum facet width shall be 4 mm on the gauge corner, 7 mm on the shoulder and 10 mm within 10 mm of the rail crown. The reprofiling zone shall be blended smoothly into the parent rail.

The maximum variation in facet width over a 100 mm length of rail shall be 25 % of the maximum width of the facet.

There shall not be continuous blueing in the reprofiling zone.



(a) Detailed section of railhead, showing reference points A and B_1 (α: angle of rail inclination)

Key

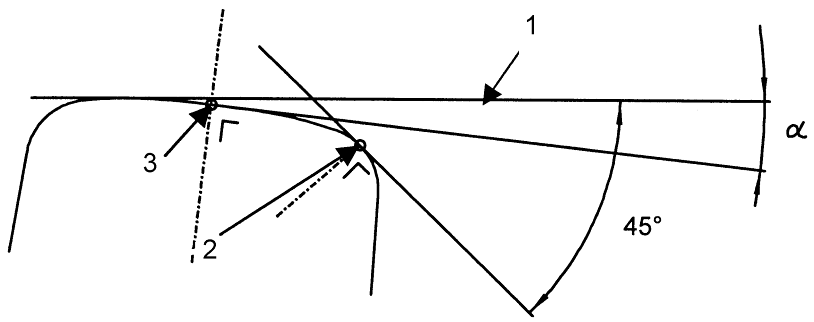
1 reference line

2 line through A parallel to reference line

3 14 mm

4 reference point B_1

5 reference point A



(b) Detailed section of railhead, showing reference point A and B_2 (α: angle of rail inclination)

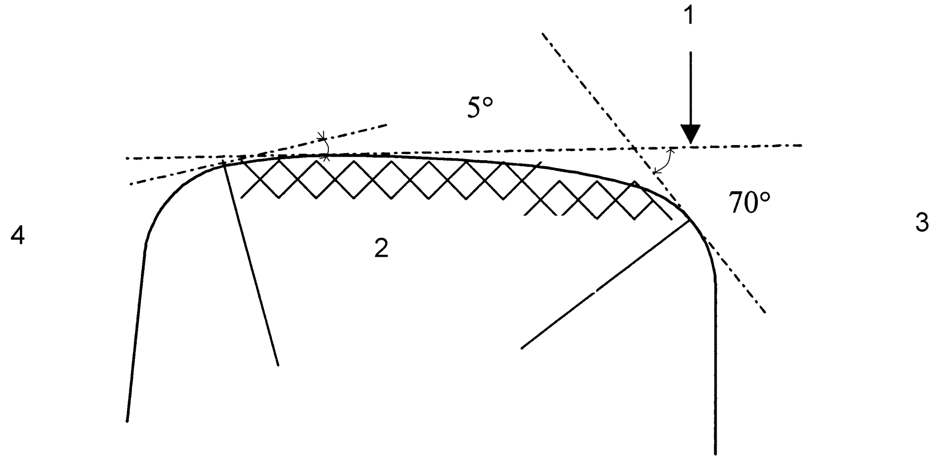
Key

1 reference line

2 reference point B_2

3 reference point A

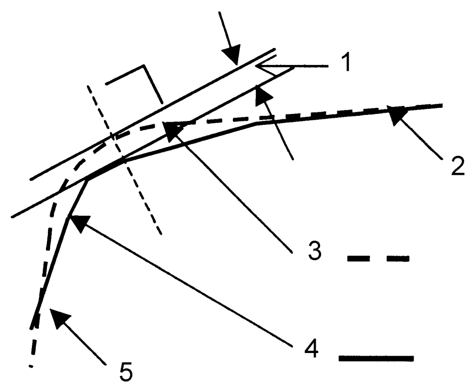
Figure 1 — Definition of terms, and determination of reference points A, B_1 and B_2 on the transverse profile



Key

1 reference line
2 re-profiling zone
3 gauge side
4 field side

Figure 2 — Re-profiling zone



NOTE In this example the range of deviation is negative (measured profile below the reference rail).

Key

1 range of deviation
2 point A
3 reference profile - - -
4 measured profile ------
5 point B_1 or B_2

Figure 3 — Deviation of measured transverse profile from reference profile

Annex A
(informative)

Procedures to verify reference instruments

A.1 Longitudinal profile

A.1.1 Principle

The longitudinal profile of a machined strip on a purpose built beam is measured using both the test instrument and a Coordinate Measuring Machine (CMM). The test instrument is assessed for its ability to make both average peak-to-peak and RMS amplitude measurements of the profile of the test strip.

The test instrument is deemed to be verified as a reference instrument for the acceptance of rails on the basis of moving average RMS values (see 4.3.2) if there is satisfactory agreement between the RMS values obtained using the test instrument and those made using the CMM. Similarly, the test instrument is deemed to be verified as a reference instrument for the acceptance of rails on the basis of moving average peak-to-peak values (see 4.3.3) if there is satisfactory agreement between the average peak-to-peak values obtained using the test instrument and those made using the CMM.

A test instrument accepted as a reference instrument for the acceptance of rails on the basis of moving average peak-to-peak values is also deemed to be verified for the acceptance of rails on the basis of the numbers of irregularities exceeding a specified amplitude (see 4.3.4).

The procedure involves the assessment of the test instrument for all wavelength ranges and all rail acceptance criteria. Acceptance of the test instrument as a reference instrument may however be limited to specified wavelength ranges and to measurements of either RMS or average peak-to-peak values.

NOTE Verification takes place at ambient temperature. Users of reference instruments are recommended to seek the guidance of the instrument manufacturer as to the temperature range over which the verification can be expected to remain valid.

A.1.2 Calibration beam

The machined strip on the calibration beam shall be not less than 20 mm wide and three times as long as the maximum wavelength for which the test instrument is to be verified.

NOTE 1 An additional running-in length at each end of the machined strip may be desirable to avoid the possibility of transient effects corrupting the beginning of a profile record.

Irregularities of the wavelengths and peak-to-peak amplitudes shown in Table A.1 shall be present on the machined strip.

Table A.1 — Calibration beam irregularities

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1 000
Peak-to-peak amplitude (mm)	0,005 – 0,015	0,005 - 0,015	0,020 - 0,040	0,080 - 0,120

NOTE 2 An example of a calibration beam suitable for the verification of instruments for wavelengths less than 300 mm is shown in Figure A.1. The longitudinal profile along the nominal centre-line of the machined strip of this beam is shown in Figure A.2. The longer wavelength irregularities can be produced by traversing the strip under a grinding head and incrementing the grinding head up and down. The shorter wavelength irregularities can be produced by linishing. The precise form of the irregularities is not critical.

A.1.3 Coordinate Measuring Machine (CMM)

The CMM shall be operated in accordance with the procedures in the manufacturer's operating manual. The maximum error, E, determined according to EN ISO 10360-2, shall not exceed [2,0 + (L/300)] µm.

A.1.4 Measurement of the calibration beam using the CMM

Clamp the calibration beam in the CMM, taking care to minimize irregularities in the beam arising from the holding arrangement.

A measuring probe with a head diameter in the range 3 mm to 6 mm shall be used.

Sufficient time shall be allowed, before measurements are taken, for electronic systems to stabilize and for the calibration beam to come to thermal equilibrium with its surroundings.

Make three records of the traced profile of the calibration beam along nominally the same path and in each direction.

A.1.5 Data analysis

Produce a primary profile from each of the traced profiles. The sampling interval shall be not greater than 1 mm and the digitisation increment not greater than 1 µm. From the primary profile produce a filtered profile for each of the wavelengths in Table A.2 for which the test instrument is to be verified. Phase correct profile filters shall be used which attenuate wavelength components in the primary profile outside the wavelength range of interest at a rate of at least 60 dB/decade.

NOTE 1 Examples of satisfactory filter characteristics are shown in Figure A.3.

For each filtered profile, the RMS amplitude, Z_{RMS}, shall be calculated as follows: if there are n individual samples z_i of the filtered profile within the measured length, then:

$$z_{\mathrm{RMS}} = \left[\sum_{i=1}^{i=n} z_i^2 /(n-1) \right]^{1/2} \tag{A.1}$$

Calculate the RMS amplitude of the combined 6 filtered profiles in each wavelength range using Equation A.2:

$$Z = \left[\sum_{j=1}^{j=6} n_j z_j^2 \Big/ \left(\sum_{j=1}^{j=6} n_j - 1 \right) \right]^{1/2} \tag{A.2}$$

where z_j is the RMS amplitude and n_j is the number of samples for the j^{th} set of filtered profiles.

For each filtered profile the average peak-to-peak amplitude shall be calculated as follows.

$$a_{\text{ave}} = \sum_{k=1}^{k=m} a_{\text{k}} / m \qquad \text{(A.3)}$$

where a_k is the amplitude of successive peak-to-peak ranges (compare Figure D.2) and m is the number of such ranges in the profile.

Calculate the average peak-to-peak amplitude of the combined 6 filtered profiles (a_{combave}) in each wavelength range using Equation A.4.

$$a_{\text{combave}} = \sum_{h=1}^{h=6} (a_{\text{h}} . n_{\text{h}}) / \sum_{h=1}^{h=6} n_{\text{h}} \qquad \text{(A.4)}$$

where, for each wavelength range, n_h is the number of peak-to-peak amplitudes of average value a_h in the h^{th} filtered profile.

NOTE 2 For a specific filtered profile (h), n_h and a_h in Equation A.4 correspond to m and a_{ave} in Equation A.3 respectively.

A.1.6 Acceptance of calibration beam

The calibration beam shall be accepted for the assessment of test instruments for the measurement of RMS amplitude if the maximum differences between the RMS amplitudes of any two filtered profiles are less than the values given in Table A.2.

The calibration beam shall be accepted for the assessment of test instruments for the measurement of average peak-to-peak amplitude if none of the average peak-to-peak amplitudes differ by more than the values given in Table A.2.

Table A.2 — Maximum differences in calculated values permitted (beam)

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1 000
Difference in RMS values (µm)	1	1	1	2
Difference in average peak-to-peak amplitude values (µm)	2	2	2	5

NOTE Acceptance may be limited to specified wavelength ranges.

A.1.7 Measurement of the calibration beam using the test instrument

Immediately following the measurement of the calibration beam using the CMM and without removing the calibration beam from the CMM, make three records of the traced profile of the calibration beam along nominally the same path and in each direction. Switch on the test instrument in sufficient time for its electronic circuits to stabilize before the measurements are taken.

A.1.8 Data analysis using the test instrument

Produce a primary profile from each of the traced profiles and from each primary profile produce a filtered profile for each of the wavelength ranges in Table A.2 for which the test instrument is to be verified, using the same algorithms and filter characteristics as would be used if the test instrument were used on railway track.

Calculate the RMS amplitude of each filtered profile in accordance with Equation A.1.

Calculate the RMS amplitude of the combined 6 filtered profiles in each wavelength range using Equation A.2.

Calculate the average peak-to-peak amplitude of each filtered profile in accordance with Equation A.3.

Calculate the average peak-to-peak amplitude of the combined 6 filtered profiles in each wavelength range using Equation A.4.

A.1.9 Acceptance criteria for reference instruments

A.1.9.1 Instruments for the measurements of moving average RMS amplitude

The maximum difference between the RMS amplitudes of any two filtered profiles, as calculated in A.1.8, shall be no greater than the values given in Table A.3a.

The maximum difference between the combined RMS for the six profiles in a specified wavelength range, estimated using the test instrument and those obtained using the CMM, shall not differ by more than the amount permitted in Table A.3b.

A.1.9.2 Instruments for the measurement of average peak-to-peak amplitude and for rail acceptance based on numbers of exceedances of specified peak-to-peak amplitudes

The maximum difference between the average peak-to-peak amplitudes of any two filtered profiles, as calculated in A.1.8, shall be no greater than the values given in Table A.3c.

The maximum difference between the combined average peak-to-peak amplitude for the six profiles in a specified wavelength range, estimated using the test instrument and that obtained using the CMM, shall not differ by more than the amount permitted in Table A.3d.

Table A.3a — Maximum permitted difference between test instrument RMS values

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1000
Maximum difference (µm)	1	1	2	5

Table A.3b — Maximum permitted difference between the combined RMS values obtained using the test instrument and CMM

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1000
Maximum Difference (µm)	2	2	4	10

Table A.3c — Maximum permitted difference between test instrument average peak-to-peak values

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1000
Maximum difference (µm)	2	2	5	10

Table A.3d — Maximum permitted difference between the combined average peak-to-peak values obtained using the test instrument and CMM

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1000
Maximum Difference (µm)	5	5	10	25

A.1.10 Test report

The following shall be recorded:

- The manufacturer and serial number or other means of identification of the CMM and the test instrument;
- The temperature of test;
- The CMM probe identification and head diameter;
- The orientation of the beam relative to the x, y and z axes of the CMM;
- The primary profiles obtained by both the CMM and the test instrument;
- The wavelength bands in which the test instrument complies with the requirements of A.1.9.1;
- The wavelength bands in which the test instrument complies with the requirements of A.1.9.2.

A.2 Transverse profile

A.2.1 Principle

Two rail sections are mounted in a calibration jig, which holds the sections at a known inclination and at a separation within the range of gauges for which a reference instrument would be used (see Figure A.4). The profile of one rail is close to a reference profile, while that of the other differs significantly from the reference profile. The transverse profile of nominally the same line across the two rail sections is measured using both the test instrument and a Coordinate Measuring Machine (CMM), the accuracy of which has been verified. The test instrument is deemed to be verified as a reference instrument for measurement of the transverse rail profile if there is satisfactory agreement between measurements made using the test instrument and the CMM.

NOTE Verification takes place at ambient temperature. Users of reference instruments are recommended to seek the guidance of the instrument manufacturer as to the temperature range over which the verification can be expected to remain valid.

A.2.2 Calibration jig

The calibration jig shall consist of two short rail lengths. They shall be held parallel to one another at nominally the same inclination, separated by a distance equal to the track gauge, and held sufficiently rigidly that they do not deflect significantly during the measurement.

The transverse profile of one of the rails shall be such that it deviates from a reference rail of choice, at an inclination of choice, by less than ± 0,3 mm throughout the re-profiling zone. The transverse profile of the other rail shall be such that it deviates from the reference rail of choice, at the inclination of choice, by at least 0,5 mm at two or more points in the re-profiling zone.

A.2.3 Coordinate measuring machine (CMM)

The CMM shall be operated in accordance with the procedures in the manufacturer's operating manual. The maximum error, E, determined according to EN ISO 10360-2, shall not exceed [5 + (L/200)] µm, where L is the nominal dimension being measured, in mm.

A.2.4 Calibration jig verification

Hold the calibration jig in the CMM, taking care to minimize any deformation that might arise from the holding arrangement. The two rails shall be measured using the CMM and a measuring probe with a head diameter in the range 3 mm to 6 mm. Sufficient time shall be allowed, before measurements are taken, for electronic systems to stabilize and for the calibration jig to come to thermal equilibrium with its surroundings.

The rails shall be shown to comply with the requirements of A.2.2.

A.2.5 Rail measurements using the test instrument

Immediately following the measurement of the rails using the CMM and without removing the calibration jig from the CMM, make three records of the traced profile of each rail. Switch on the test instrument in sufficient time for its electronic circuits to stabilize before the measurements are taken. Remove the test instrument from the rail between measurements.

A.2.6 Acceptance of test instruments

From each traced profile, both those produced by the CMM and those produced by the test instrument, create a primary profile. The sampling interval when the CMM is used shall not be greater than 2,5° throughout the re-profiling zone and the digitisation increment no more than 10 µm.

Compare the primary profiles for each rail from the test instrument measurements with that from the CMM as though the latter were a reference rail. The maximum positive and negative deviations shall be less than 0,1 mm throughout the re-profiling zone.

Compare each of the primary profiles for each rail from the test instrument one with another as though one was a reference rail.

NOTE If the three primary profiles of one rail are designated X, Y and Z, this means that X is compared with Y, Y with Z, and Z with X.

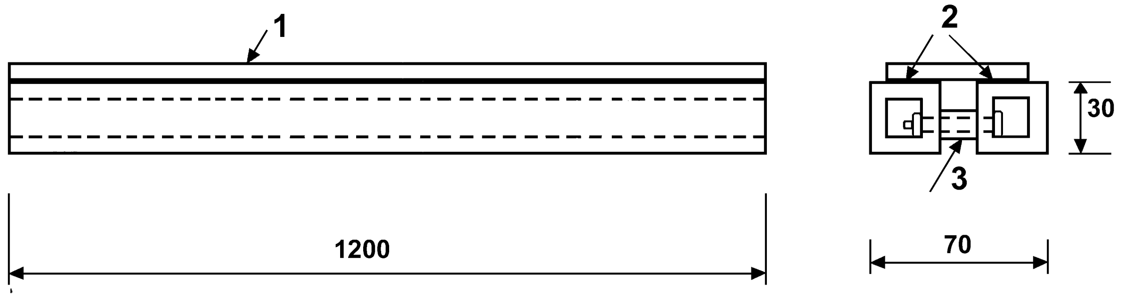
The maximum range of deviation shall be less than 0,050 mm.

A.2.7 Test report

The following shall be recorded:

- The serial number and manufacturer or other means of identification of the CMM and the test instrument;
- The temperature of test;
- The CMM probe identification and head diameter;
- The orientation of the calibration jig relative to the x, y and z axes of the CMM;
- The primary profiles obtained by both the CMM and the test instrument.

Whether the test instrument complies with the requirements of A.2.6.



NOTE 1 Dimensions are in mm and are approximate only.

NOTE 2 Profile to be measured after assembly.

NOTE 3 The beam shown is suitable for verifying instruments for wavelengths of less than 300 mm i.e. 10 mm to 30 mm, 30 mm to 100 mm and 100 mm to 300 mm.

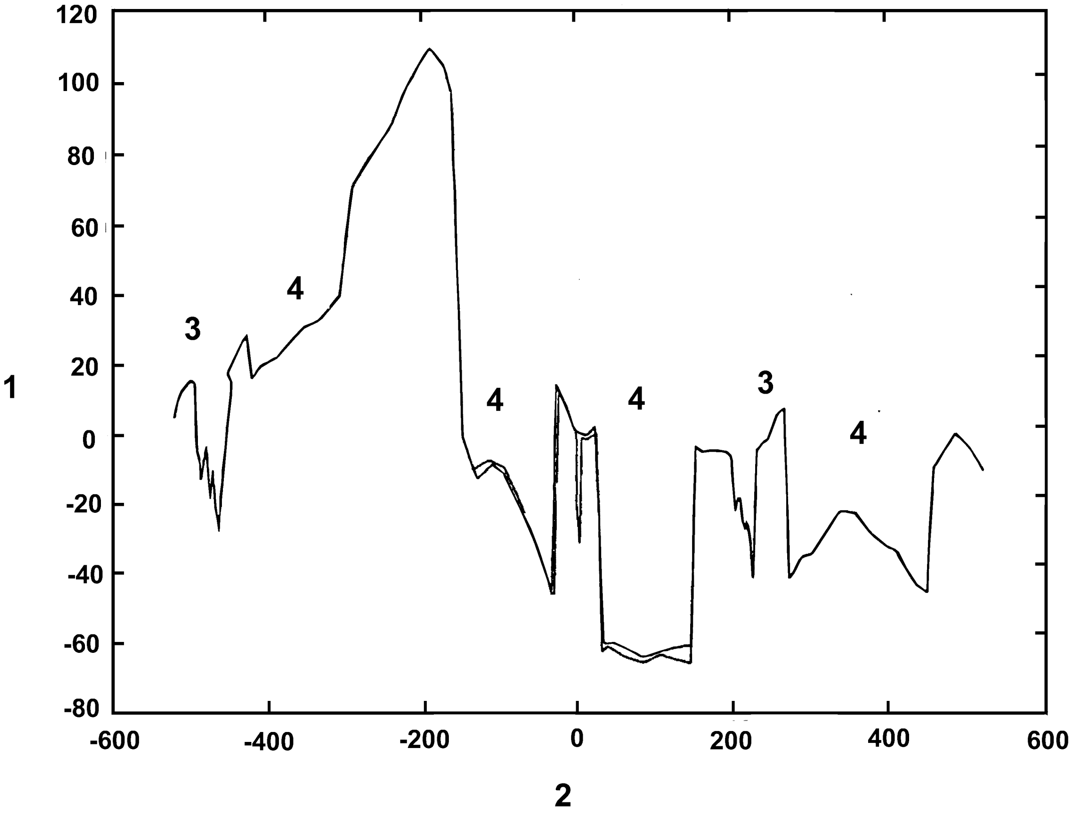
Key

1 machined strip, screwed to square section tubing

2 square section tubing

3 spacers

Figure A.1 — Schematic diagram of a calibration beam for longitudinal profile measuring equipment



Key

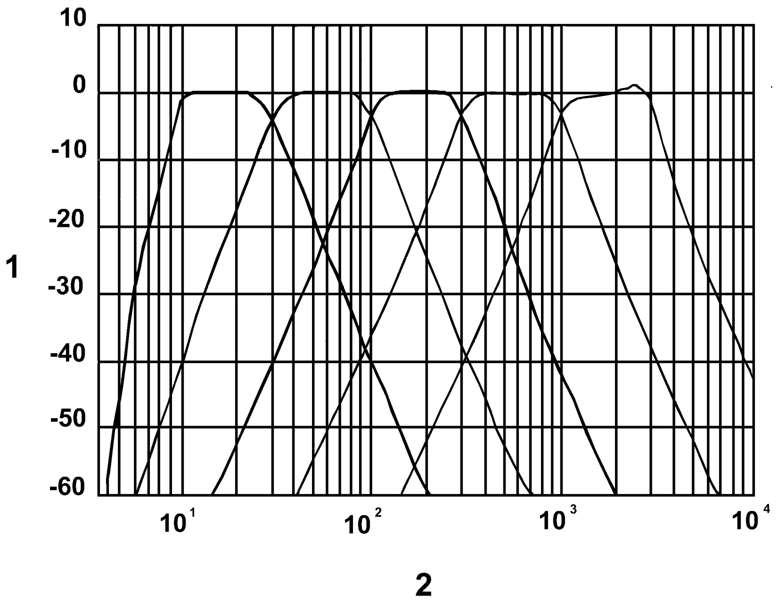
1 height (µm)

2 distance (mm)

3 linish

4 grind

Figure A.2 — Longitudinal profile of a calibration beam

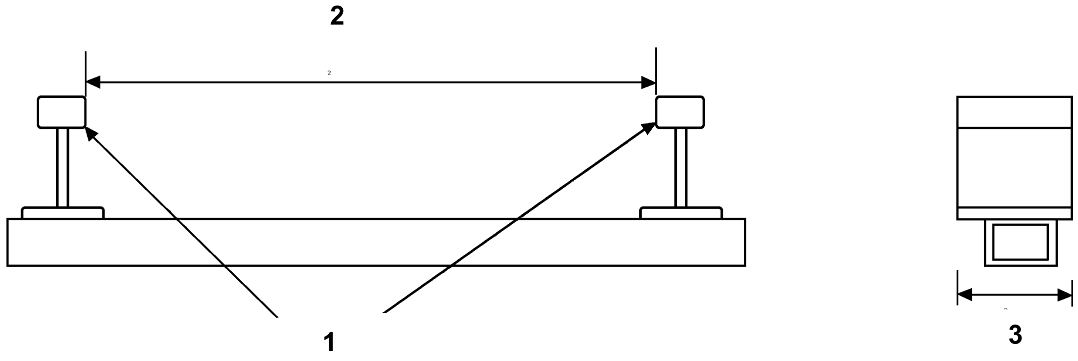


Key

1 response (dB)

2 wavelength (mm)

Figure A.3 — Amplitude response of acceptable band-pass profile filters



NOTE 1 Profile of one rail section to differ by less than ± 0,3 mm from profile of reference rail.

NOTE 2 Profile of second rail section to differ by at least 0,5 mm at one or more points in the reprofiling zone.

Key

1 rails inclined inwards at prescribed angle

2 track gauge

3 length sufficient to hold test instrument

Figure A.4 — Schematic diagram of a calibration jig for verification of reference instruments for measuring the transverse rail profile

Annex B
(informative)

Procedures to demonstrate correlation of approved and reference instruments

B.1 Longitudinal profile

B.1.1 Principle

The train-borne systems used for the routine measurement of the rail's longitudinal profile are not normally accurate enough to enable compliance with this specification to be demonstrated directly. Their use is, however, the only practical option if extensive measurements are to be made. This annex sets out the procedure to be used for the approval of systems for routine measurements.

The basis of approval is that the use of such a system shall not generally lead to the acceptance of track that would have been rejected if a reference instrument had been used and that the cumulative probability distributions for RMS or average peak-to-peak amplitude obtained using the test instrument and with a reference instrument are comparable.

To provide assurance of this, the performance of a test instrument for which approval is sought is compared to that of a reference instrument under a variety of test conditions. However if acceptance of the test instrument as an approved instrument is only required for one speed and for one direction of train travel, testing may be limited to these conditions.

The procedure given involves the assessment of the test instrument for all wavelength ranges and for the acceptance of rails on the basis of either moving average RMS (see 4.3.2) or peak-to-peak amplitudes (see 4.3.3) but approval may be limited to either RMS or average peak-to-peak values and to limited wavelength ranges.

EXAMPLE An instrument could be approved just for the 30 mm – 100 mm and 100 mm – 300 mm wavelength bands. Such an instrument would suffice to demonstrate compliance with the requirement of class 2 of 4.3.

An approved instrument for the acceptance of rails on the basis of moving average peak-to-peak values (see 4.3.3) is also deemed to be approved for the acceptance of rails on the basis of the numbers of irregularities exceeding a specified amplitude (4.3.4).

NOTE 1 The test instrument is approved on the basis of a comparison between measurements made on dry rails at ambient temperature. Users of approved instruments are recommended to seek the guidance of the instrument manufacturer as to the temperature range over which the instrument may reliably be used and whether other weather conditions could affect its performance.

NOTE 2 If the test instrument will also be used to provide information on rail condition prior to re-profiling, a similar procedure to that described here could be used to verify its suitability for that purpose but with measurements made on un-re-profiled sites.

B.1.2 Characteristics of the test sites

Three sites are required, as follows:

— (S1) Straight track

— (L1) A left hand curve of nominal radius not greater than 800 m

— (R1) A right hand curve of nominal radius not greater than 800 m

The length of each site shall not be less than 200 m.

NOTE Additional transition lengths at the beginning and the end of the measured length may be necessary.

The rails shall be dry.

All sites shall be shown to comply with the requirements of Class 1 of 4.3 using a reference instrument.

B.1.3 Measurements required

Measurements shall be taken through the test site for a continuous length of at least 200 m. If either the reference instrument or the test instrument requires a transition length at either the beginning or the end of the measurement, the transition length shall lie outside the measured length.

The longitudinal profile of both rails at each test site shall be recorded using a reference instrument.

The longitudinal profile of both rails at each test site shall be recorded using the test instrument at its maximum speed of use. Measurements shall also be made on site S1 at the minimum speed of use, if the test instrument is to be used at other than its maximum speed. If the test instrument is to be used with the train travelling both forwards and backwards, measurements shall be made with the train travelling in both directions.

B.1.4 Data analysis

The primary profiles for each rail at each site obtained using the reference instrument shall be processed, as specified in 4, to give moving average RMS amplitudes and moving average peak-to-peak amplitudes, for each wavelength range for which approval of the test instrument is sought.

The primary profiles obtained using the test instrument shall also be processed to give moving average RMS amplitudes and moving average peak-to-peak amplitudes using the electronics and software inherent in the test instrument.

NOTE 1 It is recommended that the sampling interval used by the test instrument should be no more than 2 mm and the digitisation increment be no more than 1 µm. Filters and window lengths should comply with Table 1. If a test instrument is to be used for the measurement of peak-to-peak amplitudes, the use of phase correct profile filters is expected to be necessary.

For each filtered profile obtained by the test instrument and the reference instrument, the percentage of the measured length for which the RMS and peak-to-peak moving average exceeds the values given in Tables B.1a and B.1b respectively, shall be calculated.

NOTE 2 The percentage exceedance as a function of amplitude is essentially a cumulative probability distribution for the appropriate RMS or peak-to-peak average of the filtered profile.

Table B.1a — Values of moving average of RMS amplitude for which percentage exceedance shall be calculated for each filtered profile

Wavelength range (mm)	Amplitude for which percentage exceedance shall be calculated (µm)			
10 - 30	2	4	6	8
30 - 100	2	4	6	8
100 - 300	6	12	18	24
300 - 1 000	20	40	60	80

Table B.1b — Values of moving average of peak-to-peak amplitude for which percentage exceedance shall be calculated for each filtered profile

Wavelength range (mm)	Amplitude for which percentage exceedance shall be calculated (µm)			
10 - 30	5	10	15	20
30 - 100	5	10	15	20
100 - 300	15	30	45	60
300 - 1 000	50	100	150	300

B.1.5 Acceptance criteria for approved instruments

B.1.5.1 General

An instrument can be approved for all or a limited range of wavelengths, for one or a range of speeds, for one, specified, direction of train travel or for both, for either RMS or peak-to-peak measurements or for both. Within its specified limits of use, a test instrument is required to meet all the criteria set out below for all the profiles measured.

A test instrument which has only been shown to comply with the requirements of this clause at its maximum speed of operation is deemed approved for that speed of operation only.

A test instrument which, additionally, complies at its minimum speed of operation is deemed to comply at both its maximum and minimum speeds and at all intermediate speeds.

B.1.5.2 Acceptance criteria for RMS measurements

For specified test conditions of speed and direction of travel, the percentage exceedance values obtained using the test instrument at the amplitudes given in Table B.2a for all the profiles recorded, shall not be less than those obtained using the reference instrument on the same measured lengths.

Table B.2a — Average RMS amplitudes at which test instrument percentage exceedances are not less than those obtained using a reference instrument

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1 000
Amplitude (µm)	4	4	12	40

The percentage exceedances for the test instrument for each of the amplitudes given in Table B.2a shall also be within a factor of 1 to 1,5 of the corresponding percentage exceedances obtained using the reference instrument.

B.1.5.3 Acceptance criteria for peak-to-peak measurements

For specified test conditions of speed and direction of travel, the percentage exceedance values obtained using the test instrument at the amplitudes given in Table B.2b for all the profiles recorded, shall not be less than those obtained using the reference instrument on the same measured lengths.

Table B.2b — Average peak-to-peak amplitudes at which test instrument percentage exceedances are not less than those obtained using a reference instrument

Wavelength range (mm)	10 - 30	30 - 100	100 - 300	300 - 1 000
Amplitude (µm)	10	10	30	100

The percentage exceedances for the test instrument for each of the amplitudes given in Table B.2b shall also be within a factor of 1 to 1,5 of the corresponding percentage exceedances obtained using the reference instrument.

B.1.6 Test report

The following shall be recorded:

— The serial number and manufacturer or other means of identification of the reference instrument and the test instrument.

— The nominal radii of sites L1 and R1.

— The ambient temperature and weather conditions, including the relative humidity.

— The direction of measurement with the reference instrument on each site and rail.

— The direction(s) of measurement with the test instrument on each site and rail.

— The speed(s) of measurement with the test instrument.

— The primary and filtered profiles obtained by both the test and reference instrument.

If variable, the sampling interval, digitisation increment and filter characteristics of the test instrument during its evaluation.

The speed(s), direction(s) of travel and wave-length ranges for which the requirements of B.1.5.2 are complied with by the test instrument.

The speed(s), direction(s) of travel and wave-length ranges for which the requirements of B.1.5.3 are complied with by the test instrument.

B.2 Transverse profile

B.2.1 Principle

The train-borne systems used for the routine measurement of the rail's transverse profile are not always accurate enough to enable compliance with this specification to be demonstrated directly. Their use is, however, the only practical option if extensive measurements are to be made. This annex sets out the procedure to be used for the approval of systems for routine measurements.

The basis of approval is that the use of such a system shall not generally lead to the acceptance of track that would have been rejected if a reference instrument had been used and that the percentage exceedances obtained using the test instrument and a reference instrument are comparable.

To provide assurance of this, the performance of a test instrument for which approval is sought is compared to that of a reference instrument under a variety of test conditions. However if acceptance of the test instrument as an approved instrument is only required for one speed and for one direction of train travel, testing may be limited to these conditions.

NOTE 1 The test instrument is approved on the basis of a comparison between measurements made on dry rails at ambient temperature. Users of approved instruments are recommended to seek the guidance of the instrument manufacturer as to the temperature range over which the instrument may reliably be used and whether other weather conditions could affect its performance.

NOTE 2 If the test instrument will also be used to provide information on rail condition prior to re-profiling, a similar procedure to that described here could be used to verify its suitability for that purpose but with measurements made on sites that have not been reprofiled.

B.2.2 Characteristics of the test sites

Three sites are required, as follows:

— (S2) Straight track

— (L2) A left hand curve of nominal radius not greater than 800 m

— (R2) A right hand curve of nominal radius not greater than 800 m

The length of site S2 shall not be less than 500 m. The length of sites L2 and R2 shall not be less than 100 m.

NOTE 1 Additional transition lengths at the beginning and the end of the measured length may be necessary.

The rails shall be dry.

When measured using a reference instrument, the range of deviation of all sites shall be within the limits specified in 5.3 for class Q, and the maximum deviation shall be within the limits specified for Group B, relative to the reference rail(s) of choice.

NOTE 2 Different reference rails may be used for each rail and each site.

B.2.3 Measurements required

The transverse profiles of both rails at each site shall be measured using a reference instrument and the test instrument. On site S2, measurements shall be made at equal intervals not exceeding 10 m and not less than 2 m. On sites L2 and R2, measurements shall be made at 2 m intervals.

The profiles shall be recorded using the test instrument at its maximum speed of use. Measurements shall also be made on site S2 at its minimum speed of use, if the test instrument is to be used at other than its maximum speed. If the test instrument is to be used with the train travelling both forwards and backwards, measurements shall be made with the train travelling in both directions.

B.2.4 Data analysis

Determine the maximum positive and negative deviations of each measured profile. For the reference instrument, and for each test site and rail, calculate the percentage of the measured profiles for which both the maximum positive and negative deviations fall within the following limits: ± 0,1 mm, ± 0,2 mm, ± 0,3 mm, ± 0,4 mm, + 0,5 mm, ± 0,6 mm, ± 0,7 mm, ± 0,85 mm, ± 1,0 mm. For the test instrument, these percentages shall be calculated for each test site, rail, measuring direction and speed of travel.

For each test site and rail, determine the percentage of measurements for which the range of deviation estimated by the reference instrument is less than 0,6 mm, 1,0 mm, and 1,7 mm. For the test instrument, these percentages shall be calculated for each test site, rail, measuring direction and speed of travel.

B.2.5 Acceptance criteria for approved instruments

The percentage of the profiles measured using the test instrument for which both the maximum positive and negative deviations fall within the following limits: ± 0,1 mm, ± 0,2 mm, ± 0,3 mm, ± 0,4 mm, ± 0,5 mm, ± 0,6 mm, ± 0,7 mm, ± 0,85 mm, ± 1,0 mm shall be within a factor of 0,5 to 1,5 of the percentages obtained using the reference instrument.

When compared to the results obtained using the reference instrument, the test instrument shall not overestimate the percentage of measurements for which the range of deviation is less than 0,6 mm, 1,0 mm and 1,7 mm.

A test instrument which has only been shown to comply with the requirements of this clause at its maximum speed of operation is deemed approved for that speed of operation only.

A test instrument which, additionally, complies at its minimum speed of operation is deemed to comply at both its maximum and minimum speeds and at all intermediate speeds.

B.2.6 Test report

The following shall be recorded:

- The serial number and manufacturer or other means of identification of the reference instrument and the test instrument;
- The nominal radii of sites L1 and R1;
- The ambient temperature and weather conditions, including the relative humidity;
- The direction(s) of measurement with the test instrument on each site and rail;
- The speed(s) of measurement with the test instrument;
- The primary profiles obtained by both the test and reference instrument.

The speed(s), and direction(s) of travel for which the requirements of B.2.5 are complied with by the test instrument.

Annex C
(informative)

Example of acceptance documentation for rail re-profiling work

Network ....

Region ....…………

Protocol of acceptance

Re:

Track work

Type of work: ....………
....………
Line: ....……	from km .... to km ....
Track:	from km .... to km ....
Station: ....….
Switch Nr.: ....…..
Maintenance unit: ....

Order Nr.: ....……………	dated: ....…

Contractor: ....…

Reception according (Specification Nr.: ………..)	Date: ....…

<u>Participants</u>

For railway: ....…

For contractor: ....….

Result

Work has been accepted *
Work has been accepted, but shortcomings have been listed at back-side *

Shortcomings have to be corrected until: ....
Date

Requests with regard to shortcomings possible.

* Cross off, what does not apply.

<u>Documented shortcomings</u>:

Nr.	Place	Type	Explanation	To be corrected by	Corrected Name/Date	Remark

Accepted: ...., ...
Place Date

....
Signature Representative Railway Signature Representative Contractor

---

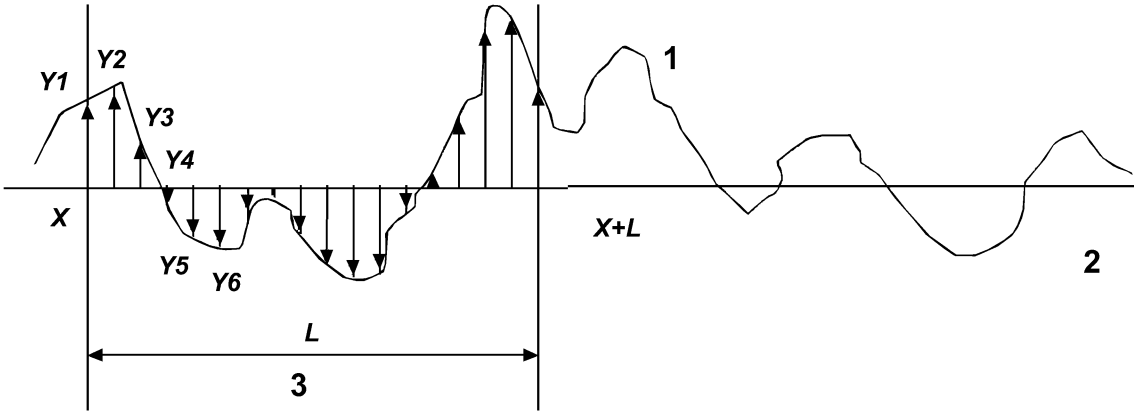
Correction of shortcomings confirmed:

...., ...
Place Date Signature Responsible Track

Annex D
(informative)

Calculation of moving average RMS and moving average peak-to-peak values

D.1 Moving average RMS values



Key

1 profile

2 line of average amplitude

3 window

Figure D.1 — Moving average RMS values

The moving average of root-mean square (RMS) amplitudes of the profile *y*(*x*) in a window of length L is in general:

$$RMS(x,L) = \left(\frac{1}{L} \int_x^{x+L} y^2(x)dx \right)^{1/2}$$

If a digitized representation of the profile is used, with values *y*(*x*), *y*(*x*+Δ*x*), *y*(*x*+2Δ*x*) etc. then equivalent RMS amplitude of the digitized profile is:

$$RMS(x,L) = \left(\sum_{i=1}^{i=n} y_i^{\,2} / (n-1) \right)^{1/2}$$

where there are n samples of the profile y within the window of length *L*.

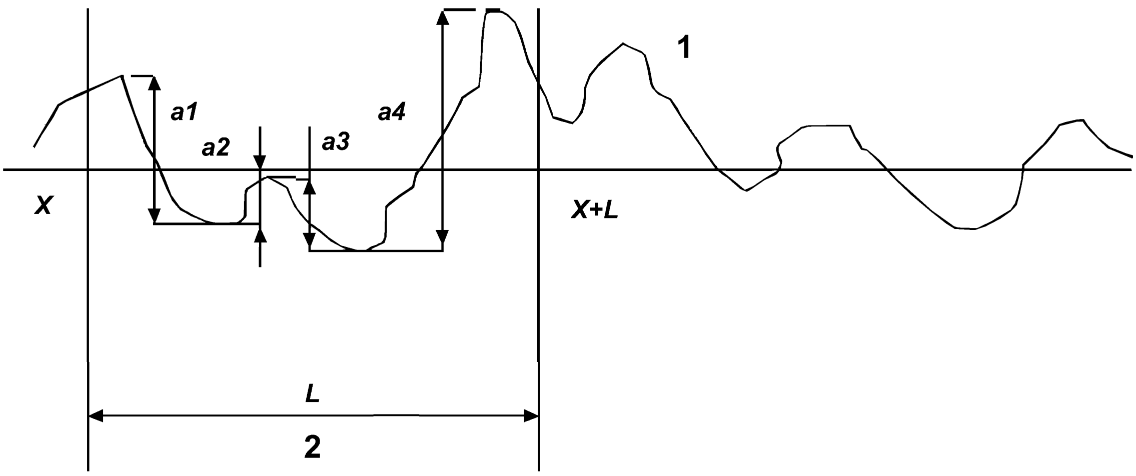
The RMS amplitude of the traced profile in a specified wavelength range (λ_1, λ_2) can be calculated as shown above by first filtering the traced profile to give the filtered profile.

The calculation may also be done from the Power Spectral Density, $S(x,L,\lambda)$, of the profile within the window of length L. The RMS amplitude in the specified wavelength range is then:

$$RMS(x,L,\lambda_1,\lambda_2) = \left(\int_{\lambda_2}^{\lambda_1} S(x,L,\lambda)dx \right)^{1/2}$$

Calculation of the Power Spectral Density of a digitized signal is a standard signal processing function.

D.2 Moving average peak-to-peak values



Key

1 profile

2 window

Figure D.2 — Moving average peak-to-peak values

The moving average of peak-to-peak amplitudes, $PP(x,L)$ within a window of length L is given by the equation:

$$PP(x,L) = (a_1+a_2+\ldots+a_n)/n$$

where there are n individual irregularities within the window of length L.

In the specific case shown:

$$PP(x,L) = (a_1+a_2+a_3+a_4)/4$$

Bibliography

[1] EN ISO 10360-2, *Geometrical Product Specifications (GPS) – Acceptance and reverification tests for coordinate measuring machines (CMM) – Part 2: CMMs used for measuring size (ISO 10360-2:2001)*